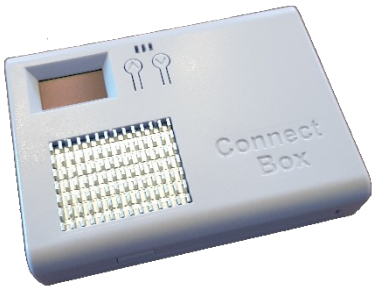
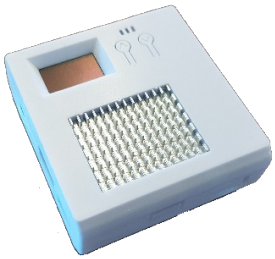
Connectbox Getting Started V11

Up to Firmware Version 20241008, plus the October and November 2026 software updates (marked NEW)







Model P V6
P/N 10001

Model S V6
P/N 10002

Model P2 V7
P/N 10011

Model S2 V7
P/N 10012

Model P3 V8 With Removable Batteries 21700 or 18650
P/N 10020

Overview and Initial Connection:

This ConnectBox is nearly ready to go, just a few things that you need to do first to make the end user experience useful.

Units shipped after May 1st, 2022 or units retrofitted with firmware version 2.0 will use this Document. Additionally, units shipped after May 1 2023 or units retrofitted with firmware 3.0 will use this document with its V3.0 notations. If your unit is older or has **Version 1.X** firmware please refer to the **ConnectBox-Getting Started_V6** document. If you would like to update to the latest firmware you may do so by downloading and burning the firmware onto a high-speed uSD card. It is important to ensure that the uSD card you use is of good quality and has the proper speed markings. The location to download the firmware is https://Github.com/ConnectBox/connectbox-pi/releases and you are looking for the latest NEO Connectbox image. This can be burned on the uSD card using a tool Balena Etcher on either a Windows machine or Macintosh machine or Linux machine. See the **ConnectBox Upgrading your firmware for hardware version V1-V3** or **V4-V8** document on the Connectbox.org website. .The platforms that are supported by version 2.0 & 3.0 firmware with additional hardware are:



Nano Pi Neo – as shown in Model P, P2 and Model S, S2 above

Raspberry Pi:

Raspberry Pi Zero W:



Raspberry Pi3b:



Raspberry Pi4b:



Raspberry Pi CM4 / CM5: \



Orange Pi:

Orange Pi Zero2:



Differences between Model P V6 and Model P2 V7 as well as Model S V6 and Model S2 V7

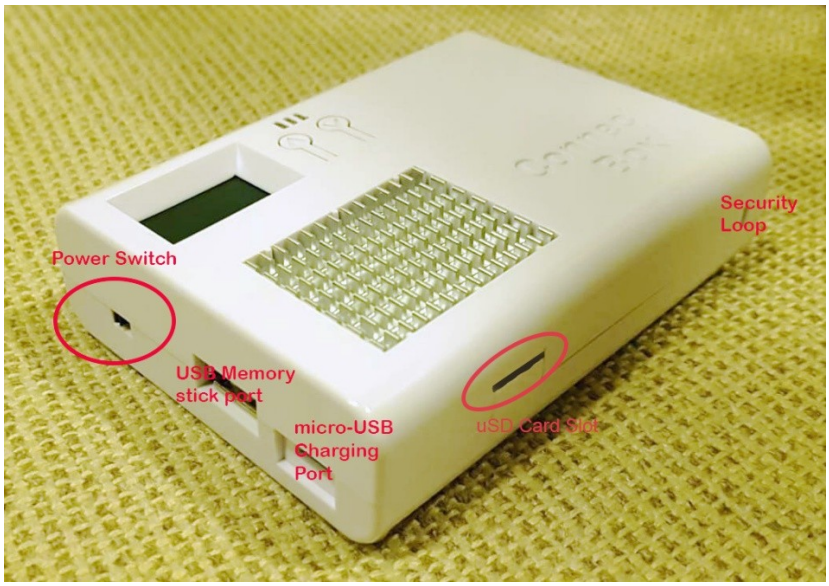
The main difference between the Model P V6 and Model S V6 vs the Model P2 V7 and Model S2 V7 is that there is a power switch on the left side panel to turn the unit on and off vs the up arrow key in the version V8 and above. Additionally there is a Wi-Fi module incorporated internally for connection to the internet via a Wi-Fi or via tethering to a mobile phone. Additionally, the Model P2 and Model S2 have OTG capability on the USB C power port allowing you to connect it directly to a host computer for data transfer or other client hardware. Also, we have upgraded the firmware from V1 – V2 allowing a more visually appealing user

experience. We have also added language detection of the client and support multilingual content based on ISO code identified folders in the data section. We have added phone home capabilities to allow a central control server to manage multiple Connectboxe's in the field receiving statistical data on usage, allowing control over all configuration features, and uploading of new content to each unique Connectbox based on location.

Another difference between a V6 hardware and V7 hardware is that V6 has a slide power switch on the side where V7 has a push for 3 second turn on and push for 6 second turn off mode on the 1st switch closest to the LCD display. The V6 switch when turned off will keep the power up on the unit for about 20 seconds before going down to allow the system to save its state. However, when you switch a unit off it will immediately stop transmission over the Wifi.

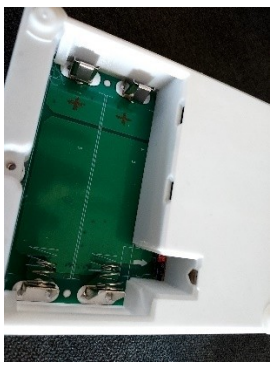
Differences between the Model P V7, P2 V7 and Model P8 is that the batteries are removable from the back of the case. Either 2 x 21700 Li-Ion batteries or 2 x 18650 with adapters can be used. The back cover also has a removable small cover for the uSD. Additionally on the side of the unit it supports a network jack for 10/100 Mbps network traffic. This can be connected to a school network or training network and act as a server for the content on the unit.

Version 6.0 hardware



Version 8.0 hardware:







Removable Batteries Removable uSD under small cover 18650's with spacing adapters

Initial Setup

1. ConnectBox comes with internal system software. No content is included other than a readme file. You'll need to add your own content to the device using the guidelines in the sections below.
2. Your battery version ConnectBox comes with an internal lithium-ion battery which is shipped at 30% charge level. If you have a Version 8 unit insert one or two 21700 or 18650 batteries with spacing adapters for the 18650's. Ensure that the positive and negative marking of the battery corresponds to the positive and negative markings on the PCB for Version 8. To charge the battery, you'll need to connect it to a 1A or greater microUSB charger like are found with many cell phones. For charging from an empty to full battery, expect it to take around 4-7 hours. Your ConnectBox has built-in low battery protection which prevents the battery from going into deep discharge as well as from overcharging.

   PLEASE FULLY CHARGE YOUR CONNECTBOX BEFORE THE FIRST USE except for model S and S2



Connectbox Model P2 V7



Connectbox Model V8

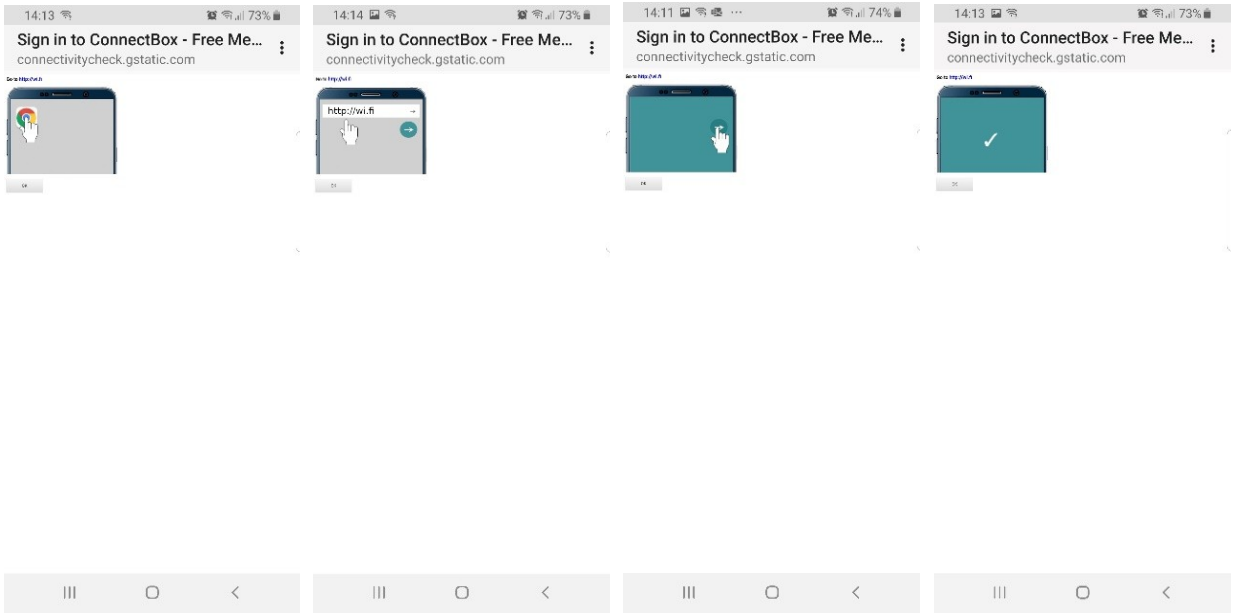
3. With the microUSB charger plugged in or with a relatively full battery, turn your unit on using the switch on the left side of the unit top. Allow a **minute or two** to pass before using a wireless device to locate and join the default wireless network with the default called *ConnectBox - Free Media*. You will know when it is available by the **orange** LED turning on. (Note: that if you charge your unit and run the unit simultaneously in a hot climate it may cause a red flashing indicator of high temperature internal to the unit.) Units' version 3- 6 of the connectbox have an on/off switch on the left side. Unit version 7 & 8 of the connectbox do not have an on/off switch. Instead hold the ^ up arrow key for 3 seconds to turn on and hold the ^up arrow key for 8-10 seconds to turn the unit off.

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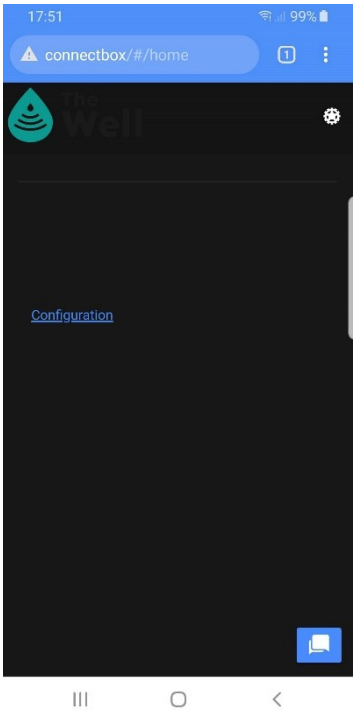
Connecting to your Device:

After the unit is on and you see the **Orange** LED indicating that the Wi-Fi hotspot is transmitting, using your mobile phone, or tablet/laptop open the Wi-Fi settings menus and look for **Connectbox - Free Media.** Select that Wi-Fi connection and your device will connect and then direct you to the captive portal to validate connectivity to the Connectbox device or check if a login is required for the network join'. In this case it will show a web browser window that say's to connect to http://gowifi.org. In some cases you may need to go to the 'more' menu (sometimes three vertical dots (⋮) and 'keep the connection'. On the captive portal page, you will be asked to enter http://gowifi in the URL window. This should open the default front page of the Connectbox. (Note: on some Samsung Galaxy phones or newer mobile phones you may be directed to turn off your cellular data for the Wi-Fi to fully work. http://gowifi.org will tell you if you need to do so. If so, turn off your cellular data connection as your phone will default to using that for internet rather than the connectbox. Once you turn off data, pull down on the screen and it will refresh returning you to http://connectbox.org or the appropriate hostname of the device.



When you're ready click OK. or use the highlighted hot web link on the top of the phone image to connect you to the ConnectBox. This is needed to make the connection and then you will be redirected back to the Wi-Fi connection page. If not, please close this browser. Once connected, you will need to open up a browser on your device (native browser, chrome, safari, Firefox, etc.) and type in the web address of http://gowifi.org which will then result in seeing the main ConnectBox window, or again if the cellular data is an issue will instruct you to turn that off. It will show a blank screen as:

NEW: The page a phone shows when it joins the ConnectBox Wi-Fi is in the phone's own language when the ConnectBox has that language (otherwise in the ConnectBox's default language), including right-to-left languages. On Android 10 and newer, if that page does not close after you press OK, tap ⋮ (top right; top left in right-to-left languages) and choose "Use this network as is".

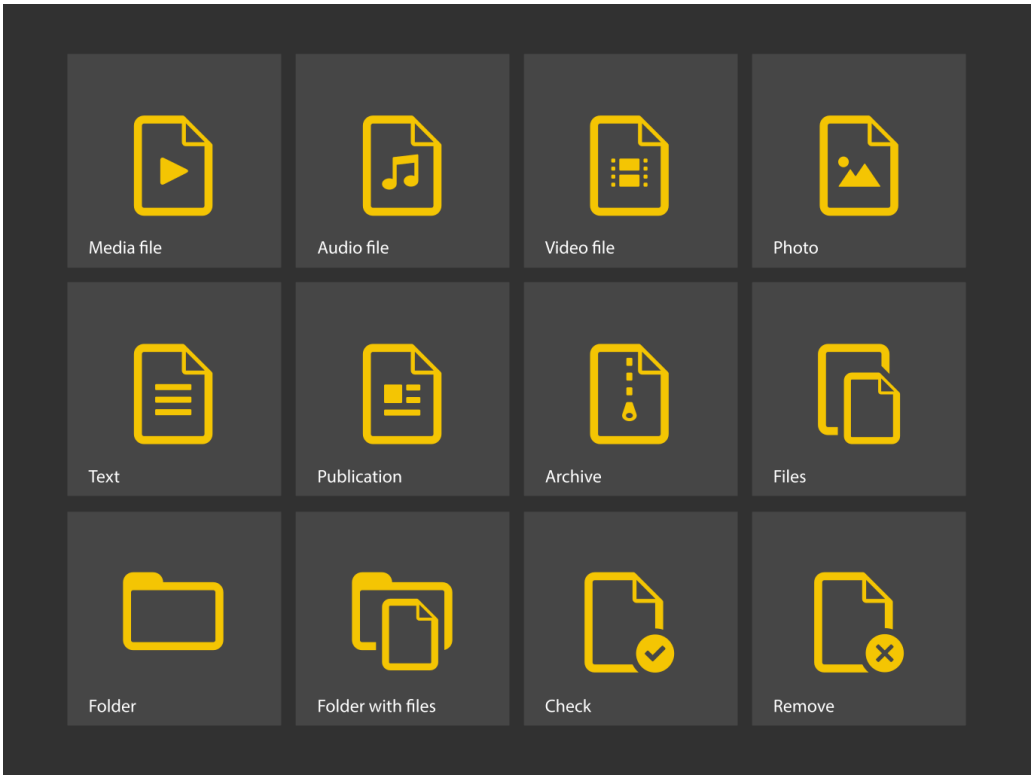


Although the default settings for the ConnectBox make it usable without any changes, data needs to be added for the users to view. Additionally, the Configuration options (discussed below) are available for those who wish to customize the operation of the ConnectBox.

Adding in Media Content:

You have the option of connecting an external USB stick to the USB A port that is available on the ConnectBox's exterior. The USB A port will in a sense override any content that is stored on the microSD card within the unit. The USB stick can be formatted as FAT32, exFAT, NTFS or EXT4 (use NTFS, exFAT or EXT4 for files larger than 4 GB; on Windows, use "Safely Remove" before pulling out an NTFS stick). We recommend a slim line USB media stick like the SanDisk Ultra Fit line for low profile and USB 3 speed is recommended for faster system operation. Disc capacities from 4 GB up to 512GB can be used. Media content can be added to it using a Windows or Mac system. Folders can be used at your choosing, and the file naming conventions can utilize the full character set of UTF-8, meaning non-roman scripts can be used. Content can also be kept on the unit itself: copy it from a USB stick with the front-panel copy function (see below). A USB stick plugged in is shown instead of the unit's own content, and when the stick is removed the unit rebuilds its menu from its own content again by itself.

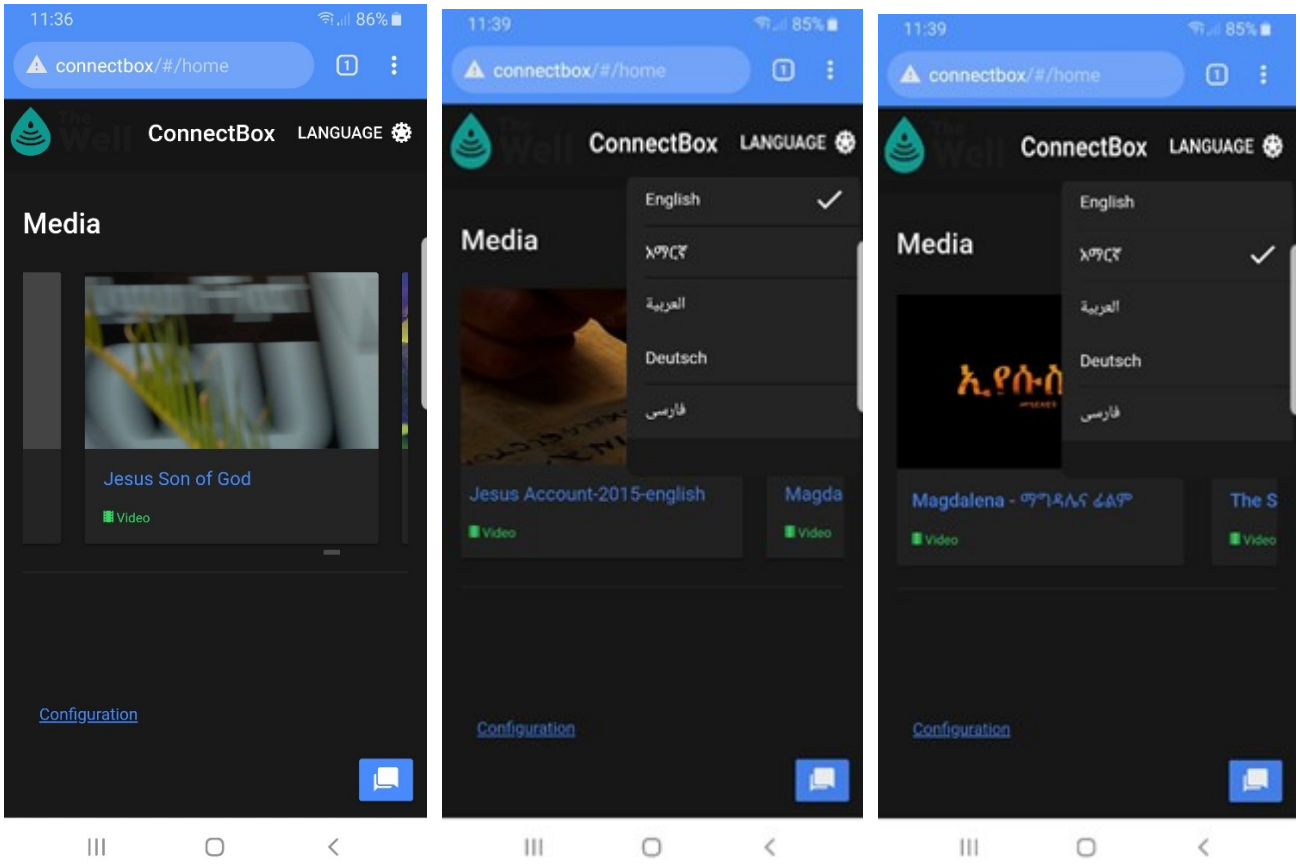
The unit will default to an enhanced user interface. This interface creates thumbnails of media content at the root level/content folder and icons of folders for specific media contents. The icons look like this:



Additionally for language support, create a folder at the content level with the ISO code of that language according to ISO 639: a 2-letter code (such as en or ar) or a 3-letter code (such as eng or ara). When a folder at the content level has a language code it will be hidden from the default language and listed in the language listing of the device. When a user with a mobile phone/tablet/computer connects to the connectbox with that language set by default in the operating system it will automatically load that language folder page as the root display. Users can always select the language they would like from the drop-down language listing: If any one language folder is used then all content must be under language folders. If no language folders are used then the contents can be contained under the root/content/ directory

NEW: The menu wording (section titles, buttons, the "Loading new content" page and the Wi-Fi welcome page) is built in for the 50 most spoken languages, so no internet is needed. Languages without a 2-letter code use their 3-letter code, for example arz (Egyptian Arabic) or pcm (Nigerian Pidgin). Chinese can be zh-CN, or a variety: cmn (Mandarin), yue (Cantonese, shown in Hong Kong Traditional characters) or nan (Min Nan,

Taiwan Traditional). See the ConnectBox Translations Guide for checking or correcting the wording.



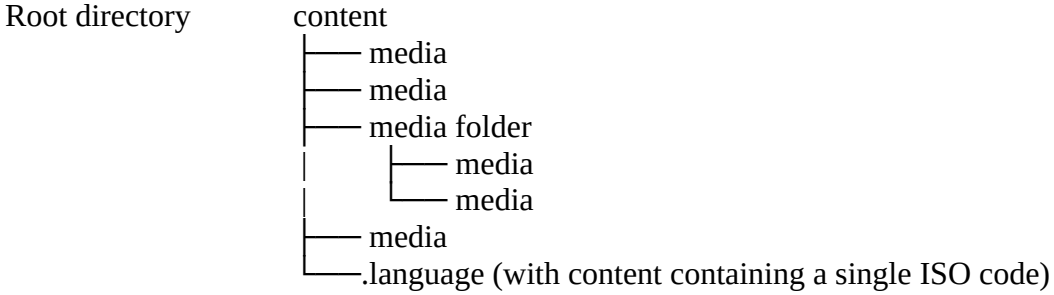
This example would have folders labeled:

root directory: ─content

- en
  - media
  - media
- am
  - media
  - media
- ar
  - media
  - media
- de
  - media
  - media
- fa
  - media
  - media

Folders at the content directory that do not contain valid ISO language codes will be ignored. There also should be no media at the content level if language folders are used as it will be ignored.

If no language folders are used then the entire contents of the USB flash drive will be considered without a selectable language. Any folders at the content level will be displayed using the default icons above with the appropriate media contents listed and any media in the content folder will be displayed as appropriate.



If you want to set the language field in the main screen you can use a file .language with the ISO code of the language of the content inside. Note that this is (dot)language under the root directory

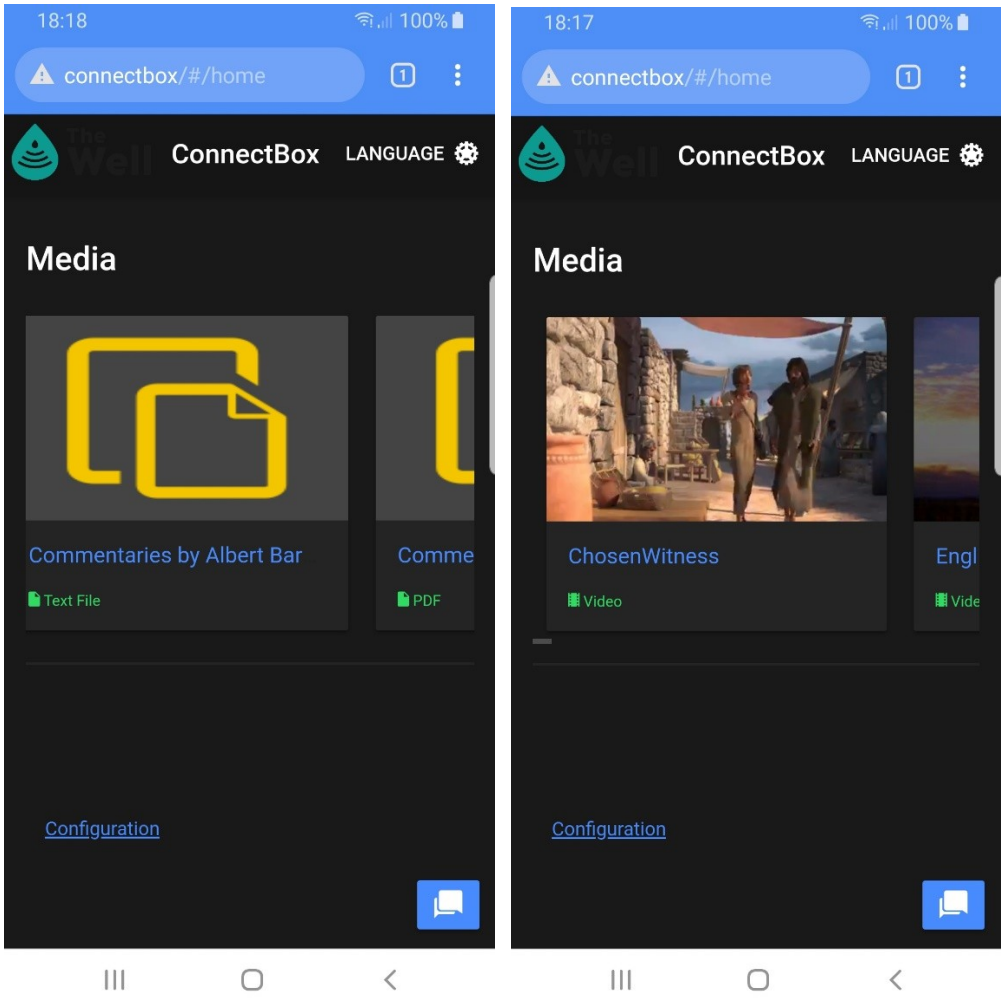
NEW: Which language the menus open in: a language chosen with the Language button always comes first (the phone remembers it). Otherwise the menus open in the phone's own language when the ConnectBox has that language, and otherwise in the ConnectBox's default language (English when the content has English, otherwise the first language).

Note: media that is at the root level or other files at the root level are ignored by the ConnectBox unless specifically used for administrative purposes. See Enabling SSH and Upgrading firmware for these cases.

Media contents are scrolled starting from right to left meaning swiping left on your phone screen will reveal additional media if available. Media that is a collection of contents will be supported under a file structure. When you enter that structure it will list the other items available. This includes audio files that are sequenced in episodes for sequential listening, chapters of books, etc. Within a media folder multiple types of media can exist. This means you can have video content along with study guides, and audio content.

Media types supported are: mp4 (video must be H.264), pdf, epub, docx (Word), xlsx and xls (Excel), pptx (PowerPoint), zim (offline websites), png, jpeg, mp3, apng, avif, gif, svg+xml, webp, webm, ogg, etc. See the detailed support documentation on browsers to get an understanding of the containers vs the codecs used in different media types.

Note: please see the File Structures Document for additional information on detailed extra files.



Note that to enable the enhanced interface thumbnails of images need to be generated to make the user experience more enjoyable. To do this when you place a new USB memory key in the unit it will mount it and then begin a process of collecting the media contents into thumbnails and screen format. While doing this you will notice on Firmware later than March 2023, a screen icon of the rotating hour glass.



This will continue until the entire contents of the USB memory stick have been processed. This may take some time due to creation of lower resolution images but will only happen once and will create a file under the

Root directory / content / saved.zip

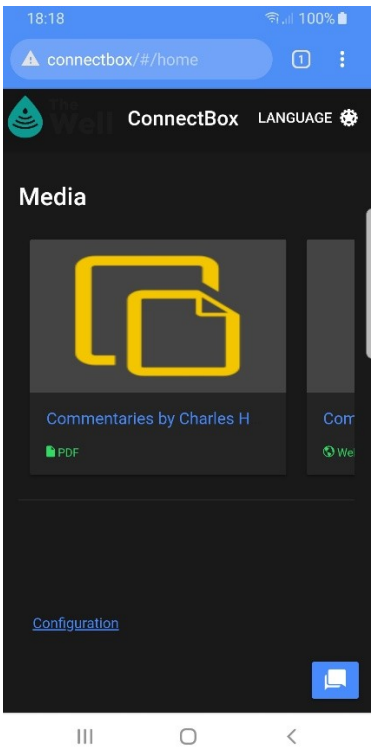
This saved.zip file contains the quick load version of the interface. It will be subsequently used to load the interface when the USB is plugged in or the unit turned on to rapidly load the interface. If you add more content to the USB memory key then you would also need to delete this one file to ensure that the new content would be included in the user menu's.

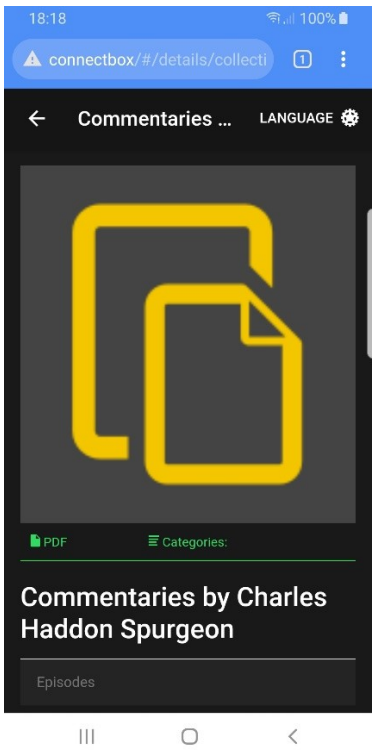
NEW: While a new USB stick is being processed, anyone who opens the ConnectBox page sees a "Loading new content" page with a progress count, in the visitor's own language when the USB has it, instead of an empty menu. It refreshes itself, and the menu appears automatically when processing is finished.

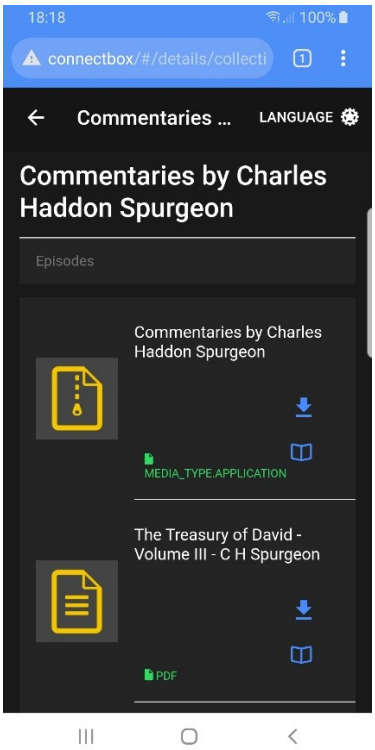
NEW: While the USB stick is being processed, the display on the front of the unit shows how far it has got, for example “Indexing 45% of 778 files”.
NEW: Before a USB stick is used, the unit may first check it for errors. The display then shows “Checking USB” and “please wait”. This happens when the stick was last taken out without being ejected – which is always the case for a stick taken out of a ConnectBox – and takes from a few seconds up to 2 minutes for a large, full stick; after that the stick is used anyway.

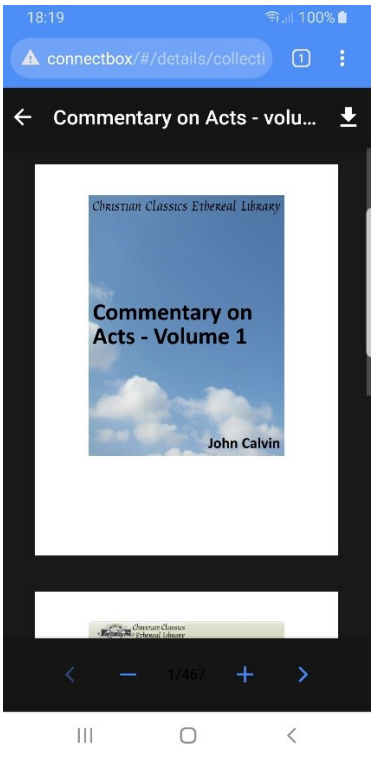
For elements of a collection:

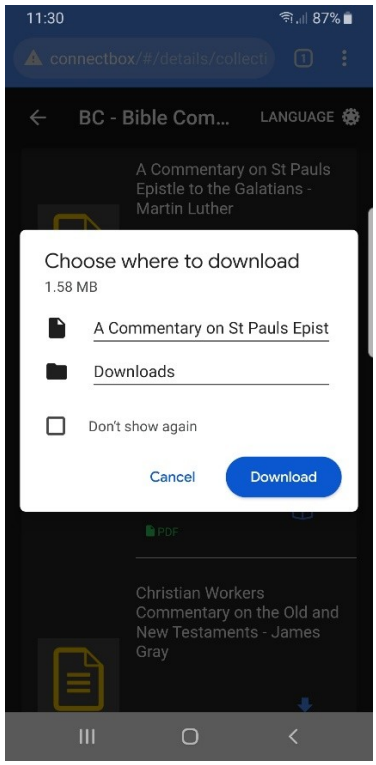
Selection of the folder then provides a view of the content scrolling up, showing the different types of content and either a view button or a download button. Selecting the view button will either show the media and allow you to interact with it or, you can select the download button to store it on your current device to look at later. Note that in this example there is a zip file of the Commentaries by Charles Haddon Spurgeon as the first item in the folder. Where the second item is a pdf file that can be opened with the built in pdf reader or downloaded.











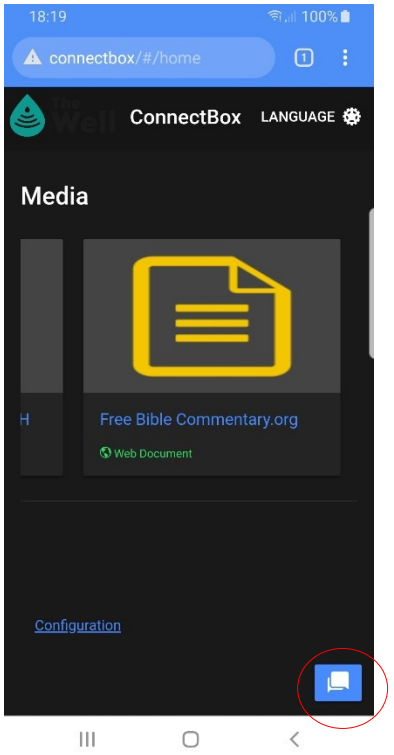
To move backwards from your current location either cancel, or use the upper left back arrow (sometimes you will find an upper right X to press to stop the media player then use the upper left back arrow. Additionally, you can always use the device back arrow button on the bottom or swipe back in some cases. In the case the file contains HTML media and an index.html file exists it will open in your browser that media directly.
NEW: a round home button in the bottom-left corner of every web page returns to the ConnectBox menu. You can also go back to the starting screen by typing in the browser address bar http://connectbox or whatever the name of your connected system is. If all else fails type http://gowifi.org in the address bar.

NOTE: That the unit has a built-in pdf reader that renders pages in HTML format as well as a built in epub reader which also renders pages in HTML. This eliminates the need for mobile devices to have built in pdf or epub readers as applications. Not however that if you download the pdf or epub then the phone must have a built-in pdf or epub reader to use offline.
NEW: Word documents (.docx) are shown as web pages, so phones can read them without an office app. Selecting one opens a details page, as for a PDF: the book button shows the document and the download button saves the original file. Text, headings, lists, tables, images and links are kept, but not the exact page layout (headers and footers, text boxes, columns), so save layout-heavy documents as PDF instead. Old .doc files are shown as web pages too (text, headings, lists and tables; no pictures), unless a PDF with the same name is in the same folder: then the PDF is shown and the .doc can be downloaded.
NEW: Excel spreadsheets (.xlsx and .xls) are shown the same way, with one table per sheet and links between the sheets. Cells show the values Excel last saved; charts, colours and number formats are not kept. Very large sheets show their first 2,000 rows, and the download button gives the whole file.
NEW: PowerPoint presentations (.pptx) are shown as one page with a numbered section for each slide: its title, text, pictures and tables, with the speaker notes underneath. Hidden slides are left out. The slide design (backgrounds, positions, colours, animations and charts) is not kept, so save design-heavy presentations as PDF instead. Old .ppt files are shown the same way (slide text and speaker notes; no pictures), unless a PDF with the same name is in the same folder: then the PDF is shown and the .ppt can be downloaded.
NEW: Every download button shows the size of the file, so users know how big it is before they download it.
NEW: Offline Websites in ZIM format (Wikipedia, Stack Exchange, PhET science simulations, TED talks and many more from https://library.kiwix.org) can be placed in a language folder. Each becomes a card that opens the website in the built-in Kiwix reader, with its own search. A ZIM file that contains several

languages is also listed in the other languages on the ConnectBox. Put each ZIM file in one language folder only (a separate “zim” folder does not work). ZIM files larger than 4 GB need an NTFS, exFAT or EXT4 USB stick.

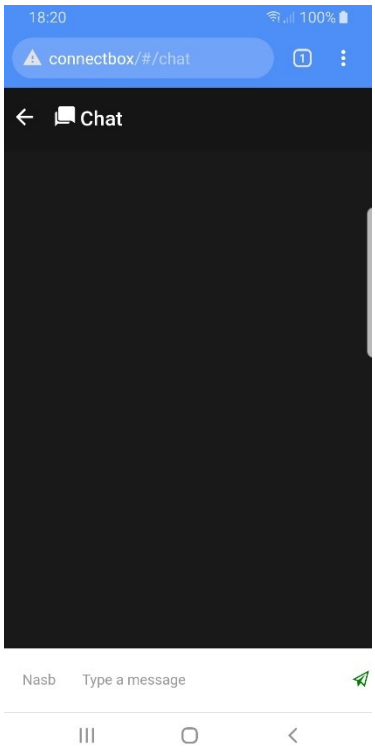
Using the Chat functionality:

Built into the main function of the Connectbox is the ability for users connected to the box to converse with one another. This is entered by tapping the chat icon on the bottom right of the screen:

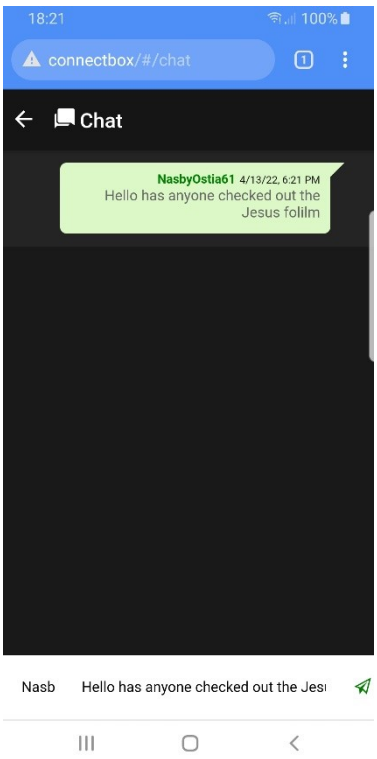


Tapping the Chat Icon

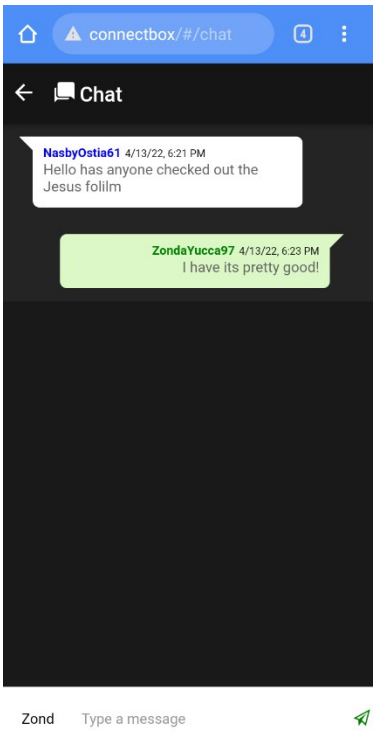
This opens the chat window:



Note that your name is automatically generated and is random on the bottom left input bar. You can change the ‘name’ of the user in the bottom left panel and enter the text of the message in the bottom right panel. Once entered it is posted by pressing the right airplane button on the bottom right the message is posted in a group chat.

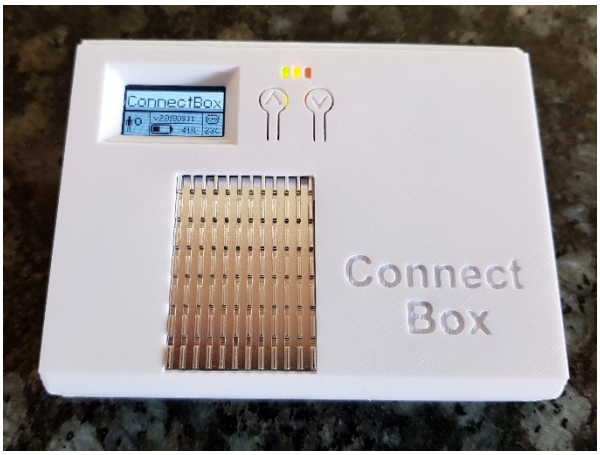


All users can see all messages that have been posted.



To exit the chat function simply press the back arrow ← □■ ♦ ♒♏ ♦ □□ ❒♓♑♒♦ □♐ ♦ ♒♏ ♦♍❒♏♏■▯

LEDs and Screen Usage:

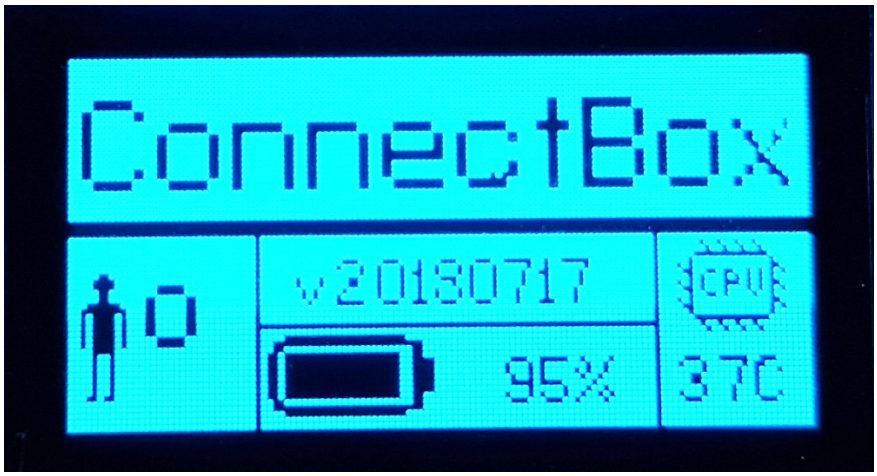


On the top front are three LED's as well as one display and two buttons. The Green LED indicates that the power is turned on. The Orange LED indicates that the system is booted Wi-Fi signal is transmitting. The Red LED is lit during battery charging and will be extinguished when the battery if fully charged. Additionally, the RED LED may blink to indicate a slow charge is occurring with the battery when the unit is charging or turned on, additionally, it may blink if the charging circuit is overheating.

NEW: The Orange LED is lit only while the Wi-Fi is really on the air: if the unit finds that its Wi-Fi has stopped, the Orange LED goes out until the Wi-Fi works again. The Orange LED also shows the battery level. It stays on when the battery is above about 63%, and about every 30 seconds it goes off for a moment once (33–63%), twice (3–33%) or three times (below 3%).

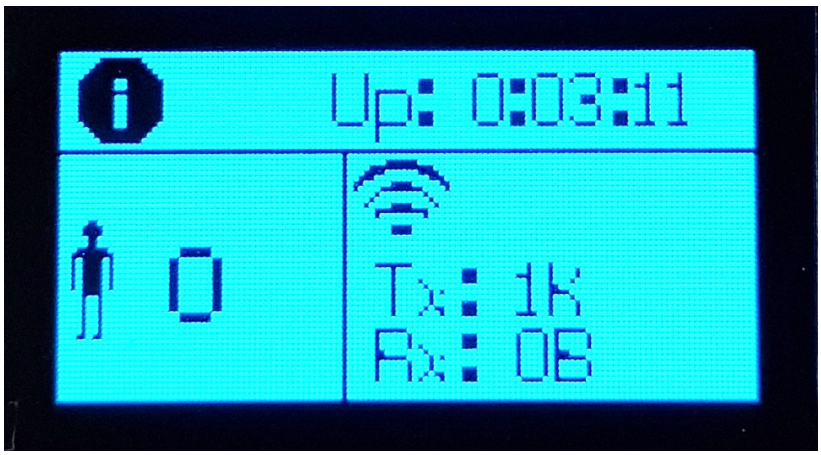
(**NOTE:** all screens can be controlled by the local system administrator and enabled or disabled. Your experience and screen movement may vary from this guide due to screens being turned off.)

The OLED screen on power up will show the ConnectBox logo for a short period of time then turn off. The two buttons, pressing the *up* button for 1 second will bring up the first status screen:



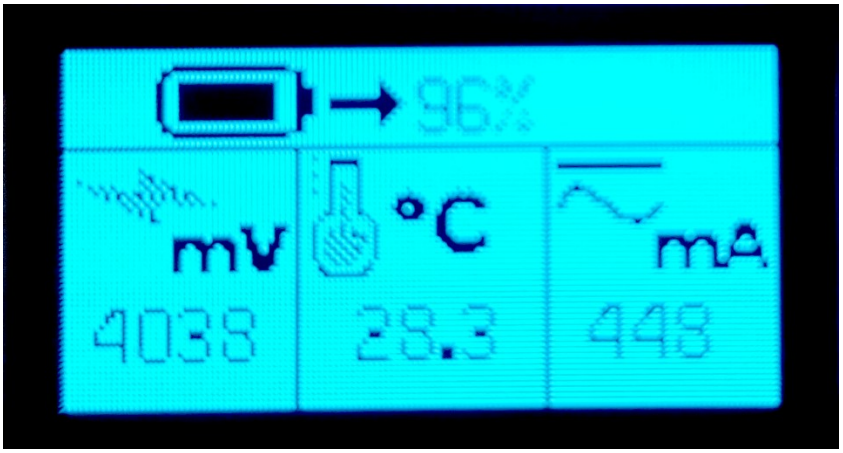
This screen shows the number of connected people, the version number of firmware in the unit, the battery charge level (Note: on a battery less ConnectBox this area is blank) and the CPU temperature. .

Pressing the up button again for 1 second with the first status screen on, will bring up the second status screen.



This screen shows the time duration that unit has been on, the number of people connected and the aggregate transmit and receive volume of data the unit has sent/received since powered on.

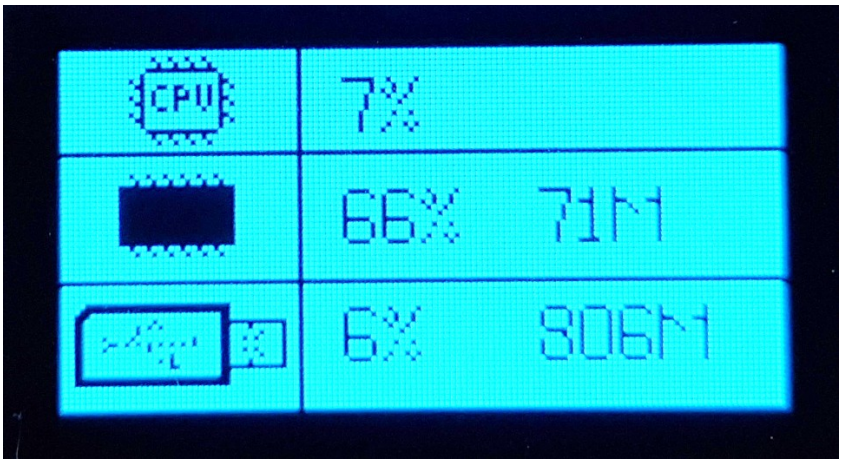
Pressing the up button again for 1 second with the second status screen on, will bring up the third status screen:



This screen shows more detailed information on the battery. This includes the current battery power level in %, the voltage of the battery, the temperature of the power control chip and the current in or out of the battery (as indicated by the directional arrow). Note that the battery is a 6400 mAh battery or a 3000 mAh battery depending on model. On a battery less unit the % area is blank and the mV area show the voltage input from the power module and current being drawn from the power module.

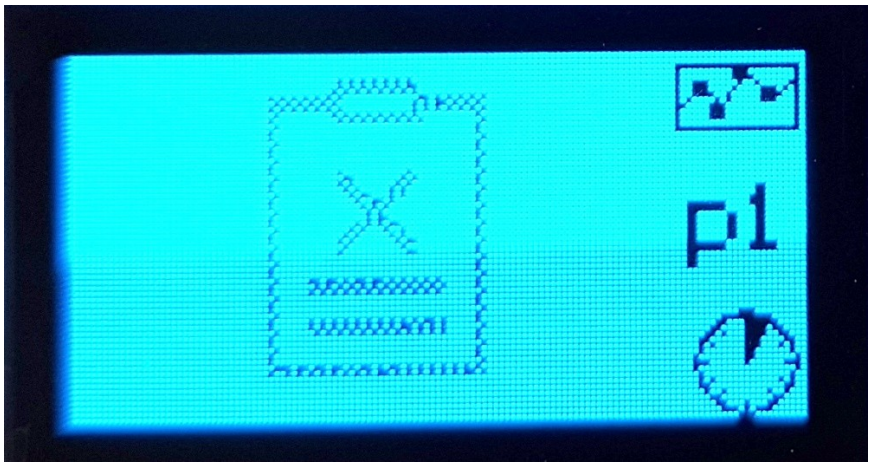
At any time after the first screen pressing the down button for 1 second will go back a screen.

Pressing the up button again for 1 second with the third status screen on, will bring up the fourth status screen:



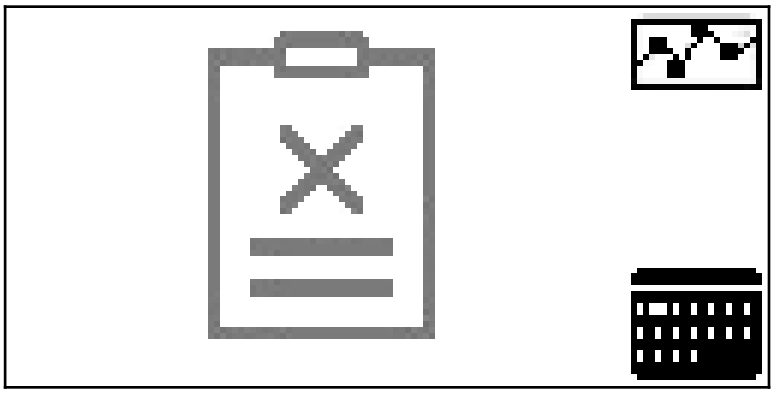
This screen shows the amount of CPU usage in %, the amount of RAM memory usage in % and the amount of USB storage used in %

Pressing the up button again for 1 second with the fourth screen on, will bring up the fifth status screen:



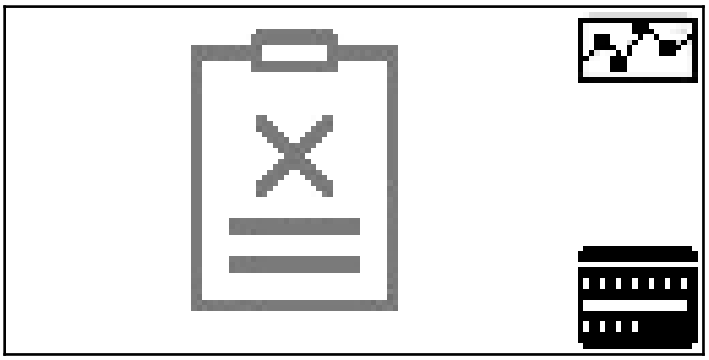
This screen shows the most popular downloaded or viewed files on the unit **over the last hour**. This screen show ‘p1’ indicating it is page one of the reports of hourly usage. Pushing the up button again to see page ‘p2’. If there is an ‘empty’ clipboard symbol, there are no statistics to view.

Pressing the up button again for 1 second with the Sixth screen up will show the Seventh screen indicating the most popular file activity for the last Day. The p number represents the page number of the report (1 or 2).

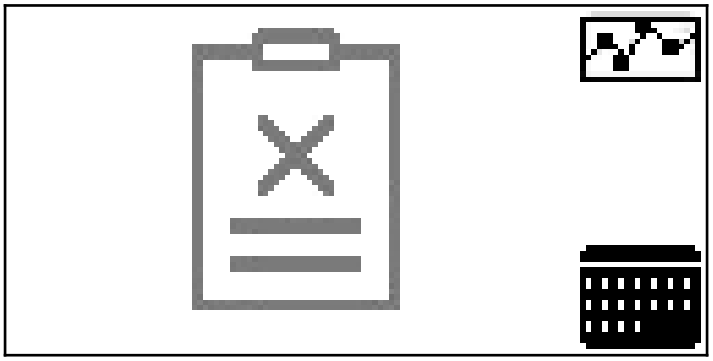


This screen shows the most popular streamed/downloaded files on the unit for the day. If there is an ‘empty’ clipboard symbol, there are no statistics to view. The p marker shows which page of the report (1 or 2) you are looking at.

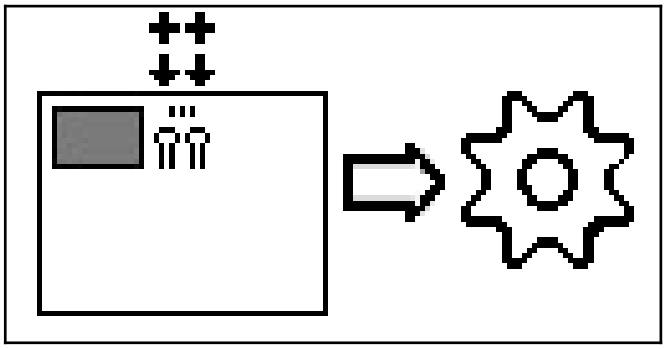
Pressing the up button again for 1 second with the Eighth screen on, will bring up the Ninth status screen showing the most popular streamed/downloaded files on the unit for the Week. If there is an ‘empty’ clipboard symbol, there are no statistics to view. The p marker shows which page of the report (1 or 2) you are looking at.



Pressing the up button again for 1 second with the Tenth screen on, will bring up the Eleventh status screen showing the most popular streamed/downloaded files on the unit for the Month. If there is an ‘empty’ clipboard symbol, there are no statistics to view. The p marker shows which page of the report (1 or 2) you are looking at.



Pressing the up button again for 1 second with the 12th screen on, will bring up the Memory copy entry screen showing how to enter file copy mode by pressing both up and down buttons simultaneously for 4 seconds.



NEW: An Internet screen comes just before the Memory copy entry screen while it is switched on in the Administration page (LCD Interface: Show LCD Internet Data Page). It shows whether sharing is ON or off and how much data the Wi-Fi users have passed through the unit: the total, Down (from the internet) and Up (to the internet). It is off until you switch it on, or until you turn on internet sharing (see Share Internet With Wi-Fi Users below), which switches it on; turning sharing off switches it off again, unless you had switched it on yourself.

NEW: An IP addresses screen comes after the statistics screens (before the Internet screen and the Memory copy entry screen). It shows the unit's three network addresses: AP is the Wi-Fi network that phones join, WiFi is the unit's own connection to another Wi-Fi network, and Eth is the network cable. A line shows "not connected" when that connection has no address, and "none" when the unit has no such connection. The WiFi address changes every time the unit starts, so read it here when you need it. It can be switched off in the Administration page (LCD Interface: Show LCD IP Addresses Page); it is on by default.



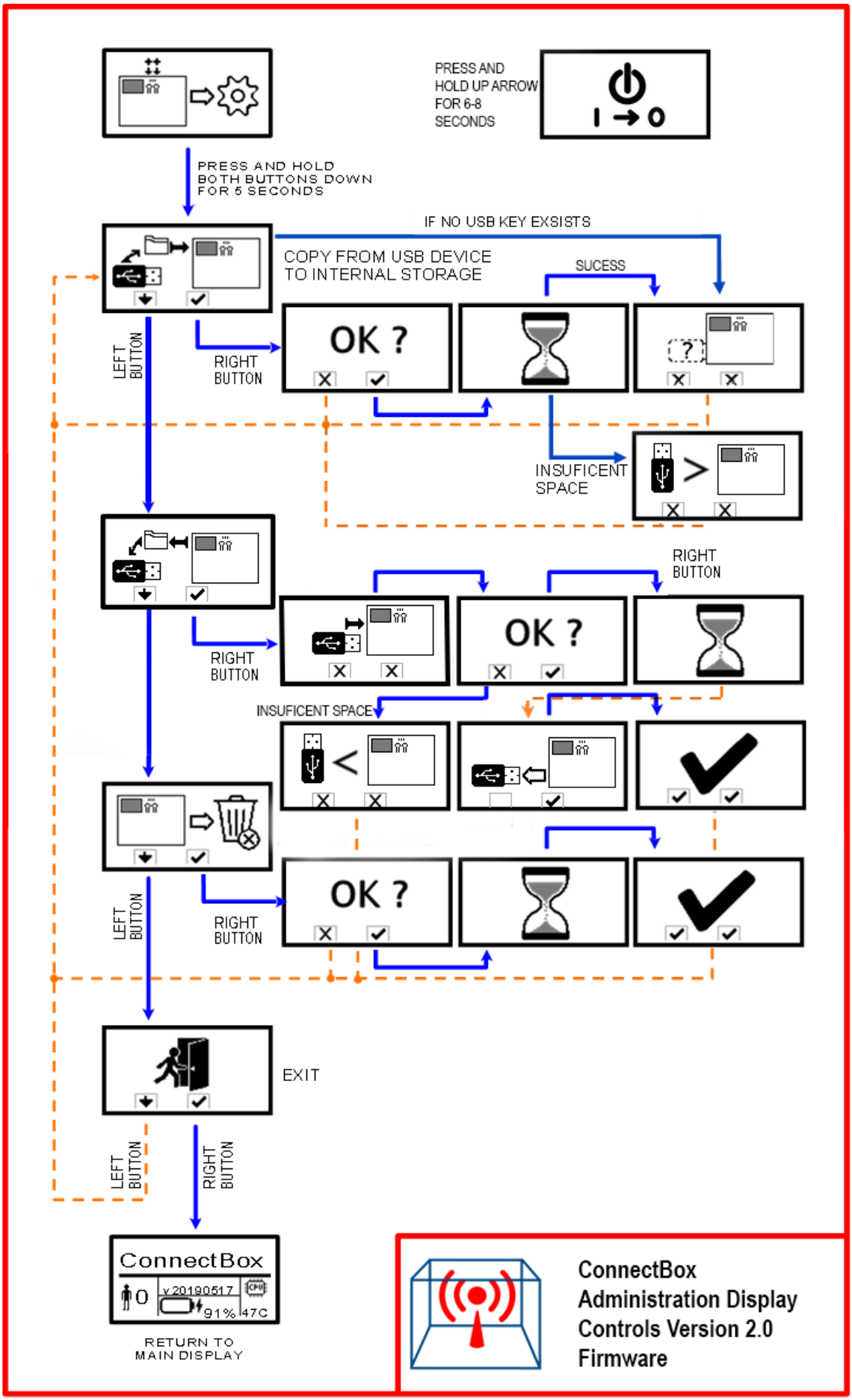
NEW: CM4 and CM5 units with the multi-battery board have an extra battery screen. It shows the battery voltage and charge, whether the batteries are charging, and the voltage of each of the four battery positions (an empty position shows 0). The unit turns this screen on by itself when it finds the battery board.

If no buttons are pressed after 10 seconds, then the screen will go black again to conserve battery power. Pressing the Up button for more than 1 second will again show the first screen.

If at any active screen you push and hold both left and right buttons for 4 seconds you will open a file copy menu which is explained in more detail in the following block diagram:

NOTE: Every screen on the front panel can be enabled or disabled from the Administration Web page. The Administrative functions menu above is disabled by default and needs to be enabled in the

Administration Web page to be visible.


PRESS AND
HOLD UP ARROW
FOR 6-8
SECONDS
PRESS AND HOLD
BOTH BUTTONS DOWN
FOR 5 SECONDS
IF NO USB KEY EXSISTS
COPY FROM USB DEVICE
TO INTERNAL STORAGE
SUCESS
OK ?
LEFT
BUTTON
RIGHT
BUTTON
INSUFICENT
SPACE
RIGHT
BUTTON
OK ?
RIGHT
BUTTON
INSUFICENT SPACE
OK ?
LEFT
BUTTON
RIGHT
BUTTON
EXIT
LEFT
BUTTON
RIGHT
BUTTON
ConnectBox
v 20190517
91%
47C
RETURN TO
MAIN DISPLAY
ConnectBox
Administration Display
Controls Version 2.0
Firmware

Once you release the two buttons you will see the first page of the administrative screen. This is a copy function from the USB key to the internal uSD memory. If you click the right check mark button (the down arrow) it will take you to the, OK? Screen double checking you want to copy from your USB key to the internal memory. If you press the checkmark button (the down arrow) it will begin to copy all contents of your USB key to the internal memory if there is enough space (your uSD has about 54GB of capacity). If there is not enough space it shows an error screen saying “Not enough room”, and nothing is copied or deleted. Otherwise, you will get an hourglass symbol as it copies the media. Be prepared to wait some time as the copy is done at USB 2.0 speeds. If you have GB of data, it may take a long time. It’s best to have the unit plugged in during this operation to ensure the battery doesn’t discharge to far during the copy. Once the copy is complete the unit rebuilds its menu from the copied content (the display shows the indexing progress) and then shows a single checkmark. Unplug the USB key and check for the content on the unit.

The second screen of the admin functions is found by pressing the left button on the copy screen reaching the Copy from Connectbox to USB memory key. This can be either a USB key or a uSD card in a USB adapter. If you press the right button (up arrow) reaching the, OK? Screen if there is a memory card installed, otherwise it will display the insert memory card display until this is completed. This OK? screen double checks that you want to copy all the data files from the Connectbox to an external memory key. During the beginning of the operation, it will double check that the memory device has enough capacity for the content. If it does it will display the OK? double check screen, otherwise it shows an error screen saying “Not enough room”, and nothing is copied or deleted.

The third screen of the admin functions is found by pressing the left button on the copy screen reaching the delete screen. This delete function is enabled by pressing the right button, the check mark. Once pressed it will ask for confirmation with an, OK? Screen. If you press the checkmark button (right down arrow) it will; proceed to delete all files on the uSD that are media. The hourglass will indicate the operation of the deletion of data files. When complete the checkmark screen will appear. Then by pressing either the right or left button you will return to the first copy screen.

NEW: While copying, the display shows the progress as a percentage and the size, for example “Copying 45%” and “Size:22431MB”. If something goes wrong, the display shows an error screen saying what (for example “No USB key”, “Not enough room”, “USB busy” or “Failed Copy”), which stays until you press a button. After a copy or delete that changes the unit’s content, the unit rebuilds its menu and shows one checkmark when it has finished. Button presses are ignored while an operation is running. Holding one button on its own for 4 seconds shuts the unit down; holding both for 4 seconds opens or closes the administrative screens.

If you are on the delete screen but do not want to delete the data on the uSD card pressing the continue button (left button) will take you to the Exit admin mode screen. If you press the continue button (left button) you will cycle back to the copy function. If you press the checkmark button (right button) you will exit the admin screens and be placed on the main page of the display

.

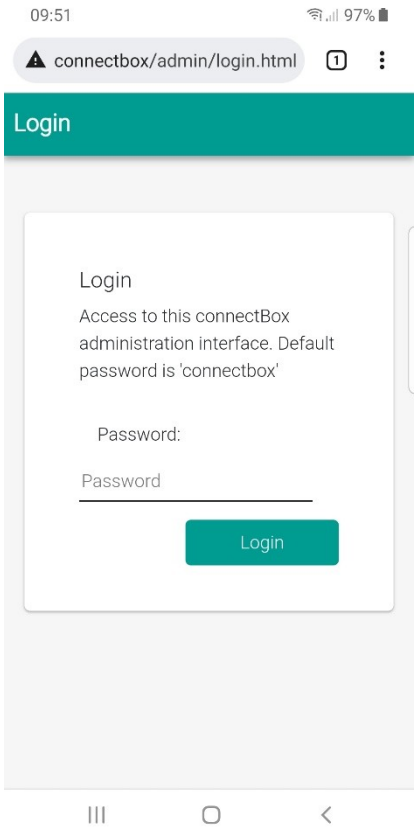
The Administrator Web Page:

Navigate to the Administration area: http://connectbox/admin (or press the cog icon on the bottom right of the screen) and login with the default Connectbox credentials (username and password are case sensitive):

username: admin
password: connectbox

Recommended: Change the password for the Administration area. Go to the Configuration Menu and select Password. Enter a new password and press submit. When you next try to enter the admin portal of the Connectbox, you will be prompted to login again. Use the new password when this happens.

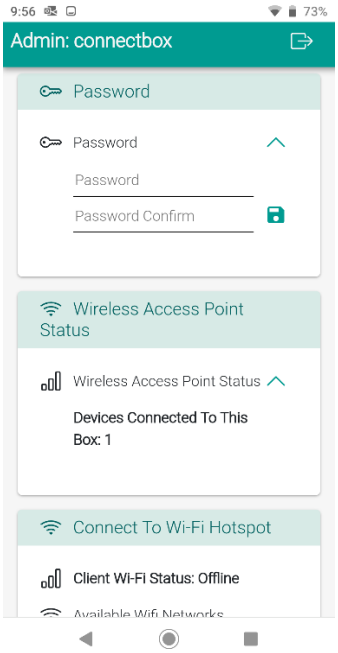
NEW: The Administration area always asks for the password. When you leave it (with Exit, the Back button, another link, or by closing the browser tab) you are logged out, and Exit takes you to the main ConnectBox screen. You are also logged out after 5 minutes without any activity on the page. This is deliberate: a phone that stays logged in as administrator can be dangerous in some countries. The login page no longer shows the default password.



Configuring your unit:

Configuration of your unit is done through the following pages:

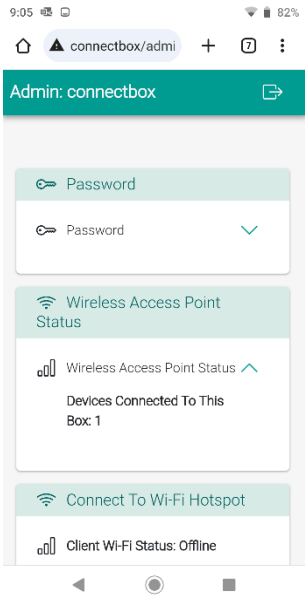
Password:



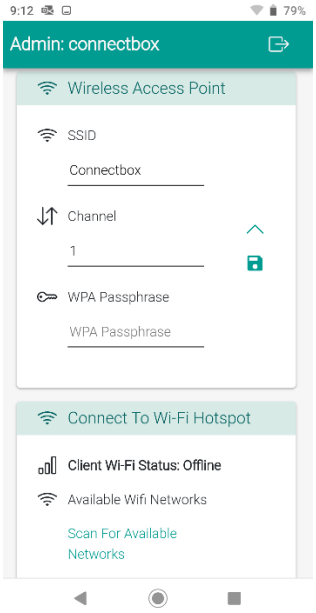
Password allows you to change the system password for the Administrator functions (Note: we highly recommend changing the administrator password from the default ‘connectbox’) to something else as soon as you get your unit. Please note that we cannot recover a forgotten or lost password.

Wireless Access Point (changes can be made only when advanced options are enabled):

The Wireless Access Point**:** change SSID (wireless display name), select the Wi-Fi channel and add Wi-Fi security by adding a WPA passphrase.



Basic Wireless Image



Advanced Options Enabled

WiFi SSID:
This is the broadcast name of the unit to clients. This can be changed to whatever you desire to identify your unit by. A maximum of 32 characters are allowed

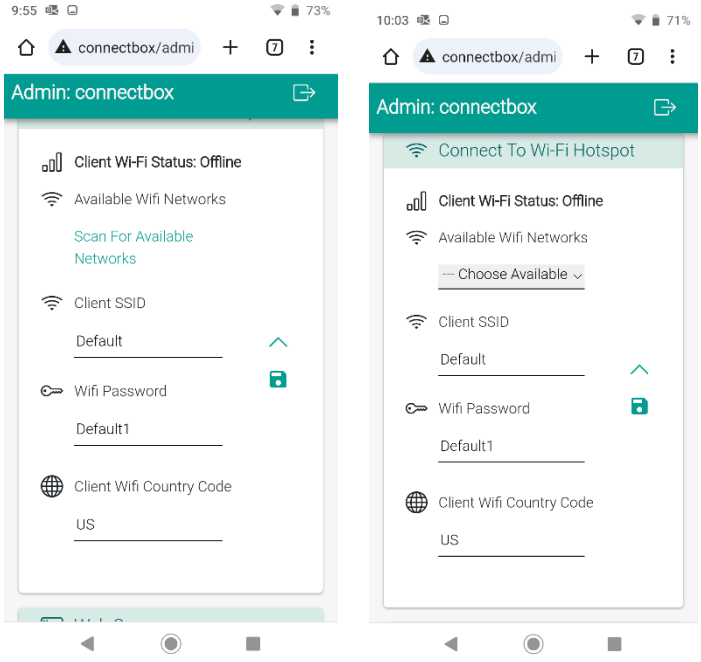
WiFi Channel:

Change the Wi-Fi channel to something that might be less congested in your area. (Configure -> Channel) Your choices are channels 1 – 11 or 1 – 14 depending on the country. It is a suggestion to allow the media box to scan all available channels in your area based on the country you're in (from the Client Wi-Fi country setting) and if appropriate will give you the least used selected channel or if another one is better it will select that one.. When you change the Wi-Fi channel, you will temporarily be disconnected form the Connectbox until your device reconnects to the Wi-Fi SSID on the new channel.

NEW: Each time the Wi-Fi starts, the unit listens to the nearby Wi-Fi networks and picks the channel with the least interference. It counts how strong each network is, and that a 2.4 GHz channel also overlaps the channels next to it (up to four channels either side), so it no longer picks a channel squeezed between two busy ones. It keeps the channel it already has unless another one is clearly better, so phones are not moved to a new channel every time the Wi-Fi restarts. A channel you set here can therefore be changed at the next start if another channel is much less busy.

WPA Passphrase:
Setting a WPA passphrase will force any user wanting to connect to your Connectbox to have a password to connect. This is appropriate in places where you want to control who connects. The Passphrase needs to be something easy to remember and should be reasonably long such as a phrase without spaces, but may contain numbers and other characters. Using good practices for creating a password is the same as creating a good Passphrase.

Client Wi-FI:
Allows you to scan and set the SSID of the access point you want to connect to internet for this device., The Wi-Fi password needed and the country code where your located to manage the channels it will look at.



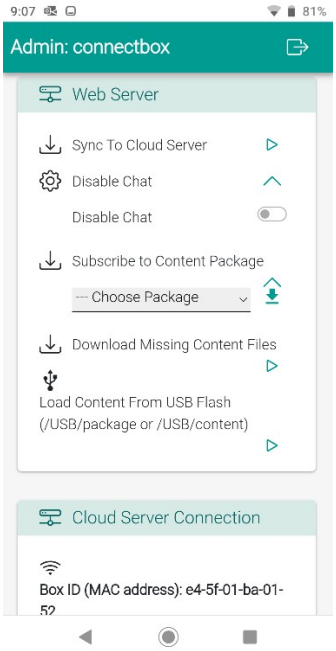
Basic Wireless Image Expanded Client SSID

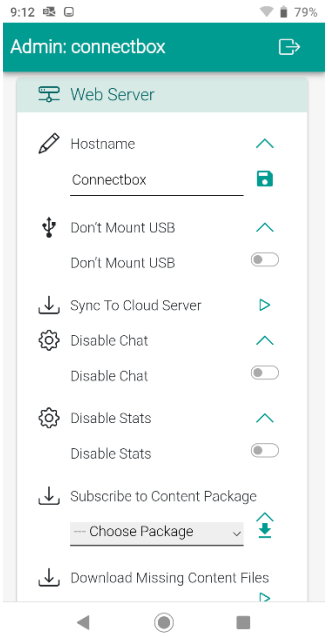
NEW: The unit remembers up to 10 Wi-Fi networks it has used to reach the internet, newest first, and connects to one of them by itself when it is in range. A network is remembered only after the unit has connected to it and reached the internet through it; when an 11th is added, the oldest is forgotten.
NEW: Wipe client SSID and password (also forgets the remembered networks): after you confirm, the unit leaves the internet Wi-Fi, the client Wi-Fi name and password go back to their default settings, and all remembered networks are forgotten.

Share Internet With Wi-Fi Users (when advanced options are enabled):
NEW: Lets phones on the ConnectBox Wi-Fi use the unit's own internet connection (its network cable or the client Wi-Fi above). It stays off unless you turn on the Share Internet switch. There is no filtering: users can reach any website. Warning: if the unit's connection is metered (mobile data, paid per MB), all of the phones' traffic, including app and system updates, counts against it. Turning sharing on or off disconnects the phones once; they join again by themselves. While sharing is on, phones do not open the ConnectBox page by themselves, so users type http://connectbox (or the unit's host name); http://wi.fi and http://gowifi.org still reach the unit too.
NEW: Data shared shows how much the Wi-Fi users have passed through the unit (down from the internet and up to the internet); the count is kept when the unit restarts. Reset data counter sets it back to 0. The front panel's Internet screen shows the same counts.

Web Server Settings (basic and advanced feature are available:

The web server can be configured to change the based Hostname that is presented to the user in the browser tab as well as options to secure the USB port to not read any device plugged into it etc.





Basic Functions Advanced Enabled Functions

Hostname:
Change the name of the hostname (by default this is 'connectbox') to something of your own choosing. From the admin menu, go to (WEB SERVER: Hostname). This only appears in the location bar of the browser. If you are changing the hostname, you will need to use the new http://hostname for the default web page, or http://hostname/admin for the administrative area. Additionally, after the change, you will have to login again as

your credentials were for the last host. After changing the hostname press the disk icon to save it and make the change permanent.

NEW: The hostname may contain only letters, digits and hyphens (no spaces, and not starting or ending with a hyphen); any other name is refused. The new name is used straight away, without restarting the unit: in the browser address (http://hostname), on the front panel and on the HDMI screen.

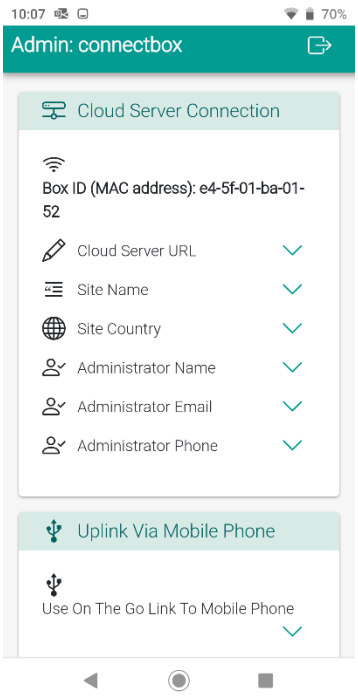
Don't mount USB**:**
This is used to lock the USB ports so that no one can insert a USB memory stick and change the content that the box is hosting.

Disable Chat:
Enabling this switch will remove the chat functionality from the unit.

Disable Stats:
Enabling this switch will stop collecting any statistics of use of content on the device. It should only be used in very sensitive situations where collection of this data could cause significant possibilities of harm.

Load Content from USB Flash (/USB/content/ or /USB/package/):
This function starts a process of checking the content and creating a quick load file as well as any thumbnail images needed for the general interface to function properly. It should be used when you have added new content or believe that the instant load file is missing (/USB/content/saved.zip)

Cloud Server Settings:
This section allows you to setup the connection to the central server for your device to report statistics and receive.



The Box ID is unique for each unit and does not change. It is the unit's wired network (Ethernet) address written with dashes, for example 02-03-5c-72-b6-f6. NEW: On NanoPi NEO units this address is set from the processor's unique serial number every time the unit starts, so the Box ID stays the same after a restart or a software re-install (the Wi-Fi addresses still change on purpose).

Cloud Server URL:
When opened will allow you to enter the cloud server name or IP address. This typically is formatted as https://chat.yourorganization.org (servers built with the current setup script use secure HTTPS). To use this feature there must be a central server setup by you or your organization to manage units in the field. This server is designed to keep the statistics of the media that has been viewed or downloaded. Additionally, it can create packages of media that will update the units in the field or alter the configuration of the field unit. Any of these features require occasional connection to the internet of the device This can be done via tethering to a mobile phone with internet, by connecting the Wi-Fi client to an internet WI-FI signal. This connection must last long enough for the unit to connect and complete its transaction. If it is downloading media, it will take longer than if just checking in and updating statistics.

Server Authorization Key (when advanced options are enabled):
NEW: The key the unit sends together with its Box ID so that the cloud server can recognise it. The unit creates it by itself the first time it reports to the server, and the server can also set it. Change it only if your server administrator asks you to.

Site Name:
The site name is an alphanumeric string that gives a reasonably readable identification of what this box is and/or where it is located. Examples may be “Cebu Bus21”. Any text is viable here. It is defined by the box administrator.

Site Country:
The site country is an alphanumeric string that gives defines the country in which this unit is operating. I provides an easy way to identify the location of the statistical data along with the Site Name.

Administrator Name is an alphanumeric string that gives the name of the site administrator for the unit. The person who has charge of the box.

Administrator Email:
This is an alphanumeric string that gives the email address of the site administrator of the box. It allows a way of communicating with the administrator on issues or questions related to the data and or media that is desired.

Administrator Phone:
This is an alphanumeric string that gives the phone number of the site administrator. Again providing a way to contact the site administrator.

Sync to Cloud Server:
This function initiates a sync either via tethering to a mobile phone or via a Client WI-FI connection and sends the statistical information about the unit to the central host server. It also looks for any new content oi content packages that need to be downloaded to update the content on this unit.

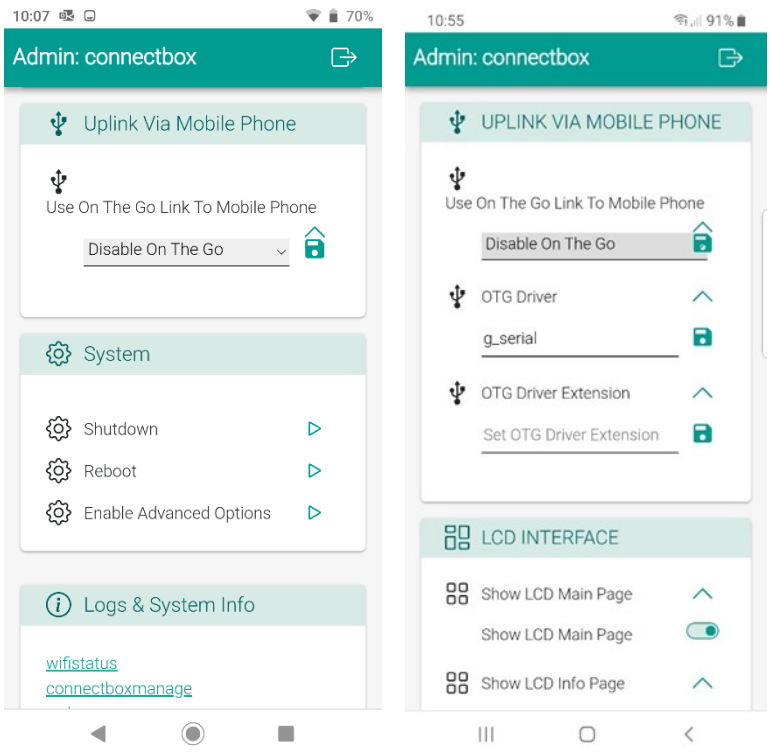
Subscribe to Content Package:
This allows you to select optional content packages from the main server to download to your device.

Download Missing Content Files:
This checks for any missing data on your unit against the server and will download any missing information to the unit. It is also how a newer version of the subscribed package is fetched: packages are not updated automatically.

NEW: Sync to Cloud Server, Subscribe to Content Package and Download Missing Content Files are now at

the end of this section (they used to be under Web Server). They appear only when the unit shows a Box ID and a Cloud Server URL has been saved.

Uplink Via Mobile Phone (when advanced options are enabled):



Basic Uplink Setting Advanced Enabled Functions

This section allows you to configure the OTG (On the Go) functionality of the device. This includes allowing the USB C port to be used as a serial port to your host computer (serial ssh into the device), or if your device will look like a hard drive to your host system, etc. Please refer to the advanced administrative guide for further information. NOTE: changing values in these fields have a direct impact on how the device functions and may cause the device to become inoperable to the Host. Only change these options when you fully understand what you are doing and when you understand the implications of what you are doing.

Note that the device is equipped with software that can detect connection to an iPhone or an Android phone automatically. So if a phone is connected by USB cable to the device it should allow for direct USB tethering without any changes in configuration.

HDMI Screen (Raspberry Pi units, when advanced options are enabled):

NEW: A monitor or TV connected to the HDMI port of a Raspberry Pi unit shows a playlist of the unit's video, audio and picture files. The sound comes out of the same HDMI cable. When there is nothing to play, the screen shows the unit's name and its Wi-Fi name.

NEW: To build the playlist, choose a file in the list and press Add to playlist; use the up, down and remove buttons beside each item to change the order. The list groups the files under headings such as Language: English, and each playlist entry shows its language in bold and underlined. Seconds per picture sets how long a picture stays on the screen.

NEW: Play playlist saves the list and starts it from the top. The playlist then repeats for ever and starts by itself every time the unit is switched on. Save playlist also saves and starts the list, unless a file started with Play now is still playing.

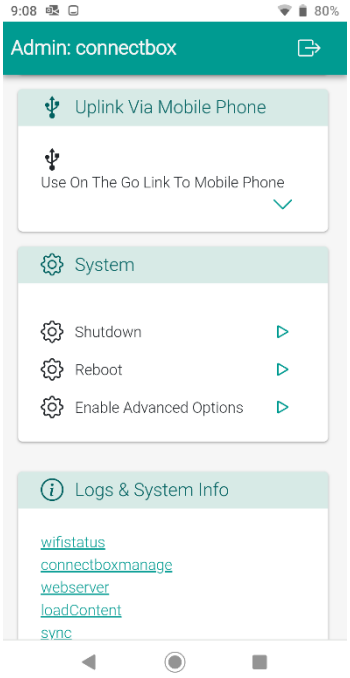
NEW: Play now plays only the file chosen in the list, once; afterwards the playlist goes on. Stop ends whatever is playing, the playlist included, and leaves the unit's name on the screen until you press Play playlist or Save playlist, or restart the unit.

NEW: Visitors can use "Play on HDMI" is a switch, off by default. When it is on, everyone on the Wi-Fi sees a screen button beside the download button of each video, audio file and picture, and can send that file to the screen. When it is off, only a logged-in administrator sees the button.

NEW: Files on a USB key are skipped while the key is out of the unit.

System:

The system menu handles the shutdown remotely or rebooting remotely of the unit. Additionally, this is where the enable of advanced options can be enabled.



Shutting Down the Unit**:**

Will shut down the ConnectBox remotely – Note: power is still on, but all functions will stop and the Wi-Fi signal will stop. To reduce battery, drain, turn the power switch off. (This can be used in creative access countries to quickly shut the unit down without having to touch the unit). Shutdown can also be initiate from the buttons on the front panel.

Rebooting the Unit**:**

Will shut down and restart the ConnectBox – Note: it will take several seconds before the Wi-Fi signal goes away and comes back again. This forces all attached devices to disconnect and then have to reconnect to use the

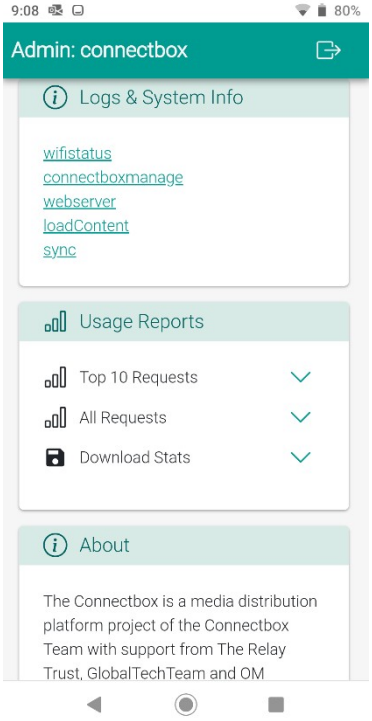
Enable Advanced Options:
This feature enables additional fields that are rarely used and are hidden to simplify the Administration menu. When you press this button, it will expand and add some menu's that are normally not there. This includes:

- Wireless Access Point Settings
- Web Server options (eg: Hostname, Don't mount USB, etc)
- Uplink options
- LCD settings of the physical unit (including advanced admin functions)
- Share Internet with Wi-Fi Users (in the Client Wi-Fi section)

HDMI Screen (Raspberry Pi units)
Box Content: copying the unit's content to or from a USB key
The browser you use remembers the choice: Advanced Options stay shown after you log out and when you log in again from the same browser, until you press Enable Advanced Options again to hide them. Each phone or computer keeps its own setting.

Logs & System Info

There are 5 different logs that can be obtained. These are used for diagnostics by support services. When you are asked to provide a specific log you can screen shot the log and send it to support.

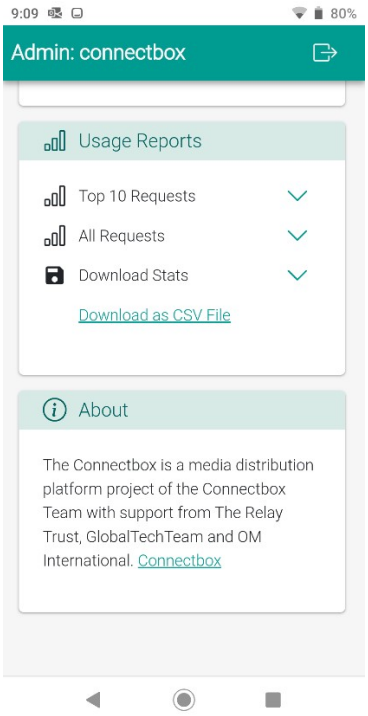


Usage Reports:

This section allows you to look at and download the top 10 open well requests from the last hour, day, week, month, or year. Additionally, all requests for the Year, Month or Week can be downloaded or viewed. These reports allow you to better understand what is important to your users and what they have been looking/downloading. These reports are part of the information that is sent to the Cloud based host package server when you connect to the Internet either by Client Wi-Fi or by tethering your unit to your mobile phone through the USB C connection.

Delete Stats:

NEW: Deletes all the usage statistics (the content viewing records) kept on the unit. It is the last item in Usage Reports (it used to be under System); download the statistics first if you want to keep them.



About:

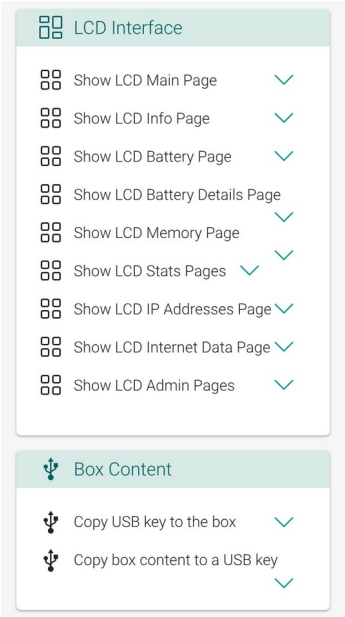
This section says who makes the ConnectBox (a media distribution platform project of the Connectbox Team), gives the email address for support, contact@globalsystems.design, and links to the ConnectBox web site.

LCD Interface (when advanced options are enabled):

This section allows you to configure what menu options are available on the front panel of the device. Each page of the device can be turned on or off depending on the values set. Note: when you set Show LCD Admin Pages to off , then all administrative pages are disabled.

NEW: Show LCD IP Addresses Page turns the IP addresses screen on or off; it is on by default.

NEW: Show LCD Internet Data Page turns the Internet screen on or off. It is off by default. Turning on Share Internet With Wi-Fi Users turns it on; turning sharing off turns it off again, unless you had turned it on yourself. A setting you made yourself, on or off, is never changed.



Box Content (when advanced options are enabled):

NEW: Copies the unit's own content (on its internal memory card) to or from a USB key, the same as the copy functions of the front panel's administrative menu; the front panel shows the same screens while it works. Open an item, press Start and confirm.

NEW: Copy USB key to the box replaces the unit's own content with the content folder of the USB key, then rebuilds the menus once the key is taken out. The unit's current content is deleted first, once the unit has checked that there is room. If the copy stops part way (for example when the key is pulled out), the part already copied stays on the unit.

NEW: Copy box content to a USB key replaces the content folder of the USB key with the unit's own content. A key formatted FAT32 cannot take a file of 4 GB or more (large ZIM files); the copy says so before it starts.

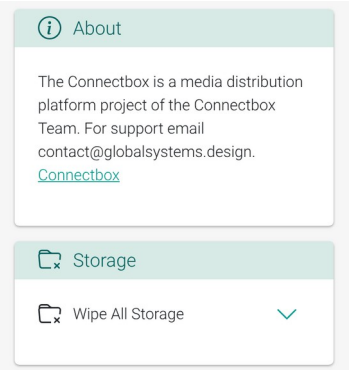
NEW: The line under the item shows each step: Put the USB key in (the unit waits up to 5 minutes), Copying 45% of 22431 MB, Copied: take the USB key out, then Done – the menus are being rebuilt, or Failed with the reason (for example No content dir, Not enough room or File over 4GB). Only one copy runs at a time, from the Administration page or the front panel. Leaving the page does not stop a copy.

Storage Management (when advanced options are enabled):

This final element of the admin menu (Wipe All Storage) is for emergencies, when the unit is found and its content could put people in danger. Enter the administration password and press the play button; with a wrong password nothing is wiped.

NEW: Within seconds the Wi-Fi goes off and stays off, every USB key in the unit is made unreadable, and on the unit's internal memory card the media and everything that names or shows it is deleted: the unit's own content, the menus and pictures, the ZIM list, the HDMI playlist, the chat, all logs including the usage statistics, and the swap area. Then, for as long as the unit has power, the USB keys and the free space of the memory card are overwritten with random data again and again; a USB key plugged in during this is destroyed too.

NEW: The unit's own software is kept: after the power is cut and the unit is switched on again, it works as before, with empty menus and the Wi-Fi on. To make it useful again, load content from a USB key or from the Cloud based host package server. A wiped USB key must be formatted again before it can be used.



Enabling SSH for the device:

There is a way to enable SSH into the unit on boot. This is accomplished by plugging in a USB Flash drive with the following structure:

root directory: ─.connectbox
└── enable-ssh

Note that the directory is (dot) connectbox with a single file enable-ssh that can be blank or empty.

As soon as this is plugged in to the device it will activate the SSH for the root account.

Upgrading the firmware with interim patches:

It is possible that interim patches will be released for V2+ firmware. These patches are applied in the following manner. Using a clean USB flash drive load the patch code into the following directory structure:

root directory: ─.connectbox - upgrade

```
├── Prerun
├── Postrun
├── update
├── CM4version
└── upgrade.py
```

Note again that this directory is (dot)connectbox/upgrade. When this USB flash drive is inserted, the operating system will detect the upgrade folder and look for the upgrade.py file. If it exists, it will then copy the upgrade code/directories to the internal memory and execute the upgrade. If successful it will then create an Upgrade xxxzxxxx.log file in the uSD directory /usr/local/connectbox/ . Additionally, it will delete the folders and upgrade.py file from the /tmp directory. If the upgrade fails, you will get a message on the display indicating failure and an error code. Contact help@connectbox.org for additional assistance. If the upgrade was successful, you will receive a notice of success on the screen and you can remove and reformat the USB flash drive you used.

Enabling Custom Web Interface:

By default, your ConnectBox is configured to run using what we call the "Enhanced User Interface". In this mode, the content that is displayed in your end user's browser is either a single piece of media which has a thumbnail generated for it (such as a frame from a video) or an Icon indicating the type of media it is along with a name. If you need to host a unique website for your application, please refer to **Creating a unique website configuration document**. This document explains how to create a unique website or to install other web tools such as WordPress.

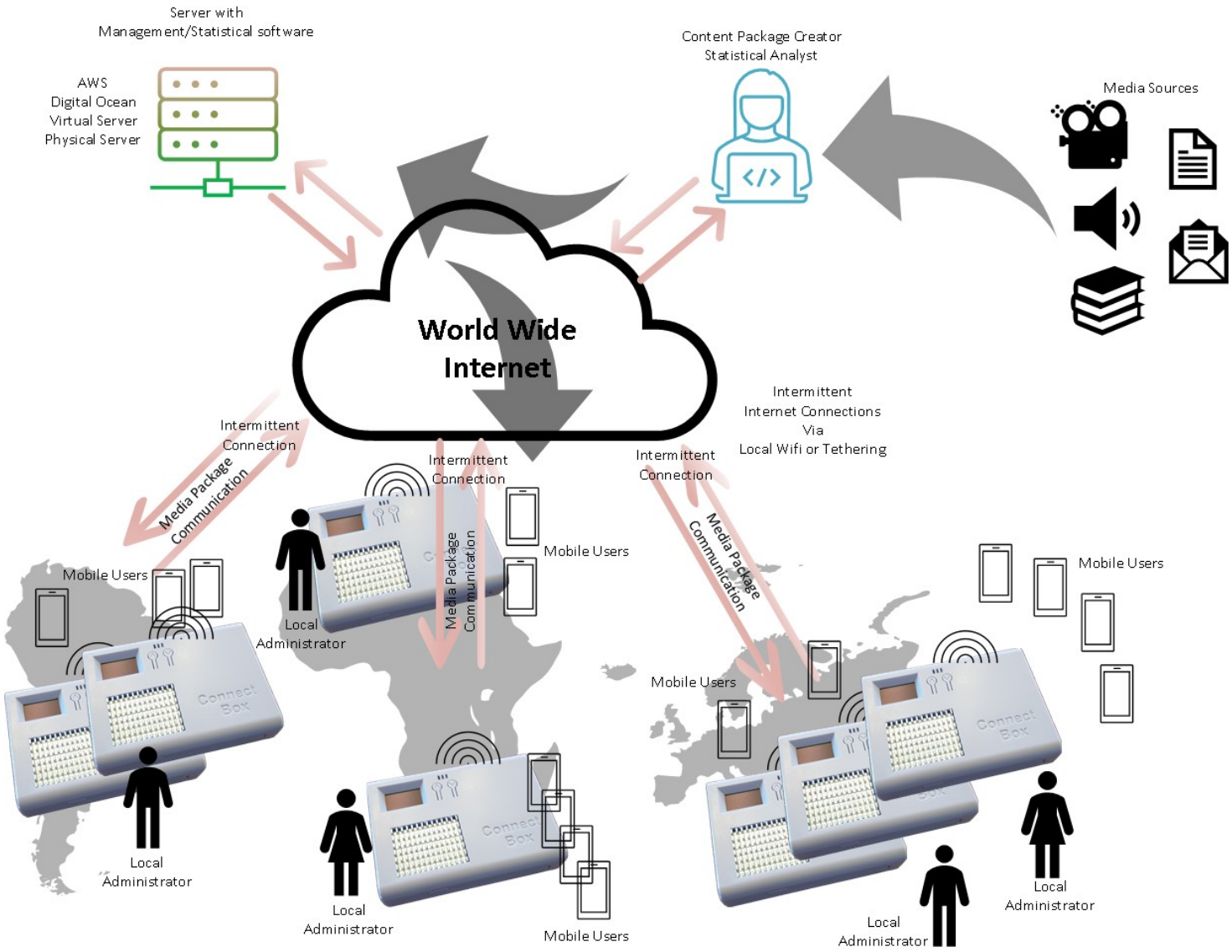
Enabling Custom WordPress installation:

If you desire to create your own web interface using the NGINX server, you can do so by following the **Configuring a ConnectBox for WordPress document**.

Linking your Connectbox to Control Host:

The connectbox can be run independently in the field or it can be connected periodically to a control host system for an organization or for reporting of statistical information for media being used from another vendor. The host management system allows for content packages to be prepared for download to specific Connectboxe's. It also allows for return uploading of statistical information from the connectbox in the field. This is done by connecting the connectbox to the internet periodically either through an access point with internet access, or through a mobile phone with a data plan tethered to the connectbox. (Note an access point can simply be a local sharing of your mobile data plan as a hotspot for the connectbox to connect to.)

The following diagram outlines the general approach for remote Connectboxe's being integrated into the Control Host:



Here media is prepared by the content package creator through the package manager interface of the Control Host system. These packages are then queued for download to the individual Connectboxe's based on unique ID of the box and location/local administrator. When the local connectbox has an internet connection either through an Access Point (AP) or by the local administrator connecting it to one, the media package will be downloaded and installed. Simultaneously it will have its media usage statistics sent to the Control Host system. The Content package creator/or Statistical analyst can then aggregate the information of media usage across units to provide reporting to media vendors that created the content. There are specific role responsibilities in this model.

1) Local Administrator, their job is to secure the unit, manage the units' capabilities locally, connect it periodically to the internet for retrieval of new content as well as reporting of content usage.
2) The Media content creator job is to prepare the packages of media that each unique unit needs as well as to oversee the ensuring of periodic reporting in of the individual boxes from the field through the return of data from the units.
3) The Statistician who's job it is to aggregate the statistical information from the Host Controller system and report back to the media providers and internally to the organization.
4) The System Administrator's job is to set up and manage the Control Host System. This is done either virtually on an Amazon Web Server or other hosting provider, or locally on physical hardware. They are to ensure the ongoing maintenance and patching of software to ensure security of the system.

To facilitate the deployment of the Connectboxe's for an organization, we provide an Ansible setup script that installs the Host Controller (chathost) and the content package builder (MediaBuilder) on an Ubuntu 24.04 server from any provider (AWS, Digital Ocean or your own hardware), with secure HTTPS access set up automatically. DNS records for chat.<your domain> and bolt.<your domain> must point to the server. See the connectbox-chathost project on GitHub.

Please refer to the documentation on setup and configuration of the control system as well as management and use of the content creator and statistical reporting sides. This will also then provide the information needed in setting up the individual ConnectBoxes.

Software versioning vs Hardware Versioning

There are both hardware versioning which this document discusses as V6 – V8 and represents versions of hardware different from one another. You will see the exact details of the differences in the Specifications below.

The software version is date based and is displayed on the main front panel of the unit. On the main screen just above the battery symbol.



As in this case v20180717. If you are running on a different hardware set than the Connectbox. The firmware revision is listed in a file on the uSD card in the FAT partition as ConnectBox_version

Features:

The Connectbox can be ordered in five models:

Battery Version with 6,400 mAH battery	Model P V6, Model P V7
Non-Battery Version with power module only	Model S V6, Model S V7
Battery Version with removable batteries	Model R V8

Model P version is larger and a bit heavier due to the batteries:

Version 6 & 7

109mm L x 80mm W x 28mm H

208 g Weight

Model P version 8 without batteries:

114mm L x 95mm W x 32mm H

172 g Weight (+2 x 21700 batteries 310g Weight)

(+2 x 18650 batteries w/ spacers 270g Weight)

Model V6/V7 S version is smaller due to the lack of the battery:

73mm L x 80mm W x 28mm H

99 g Weight

The P and S versions can be ordered with two accessories:

Accessory 1:

Metal tie wrap to attach the Connectbox to another object/anchor to keep it from being removed. Once attached the only way to detach it is to cut the metal tie-wrap.

Accessory 2:

Micro-SD card cover that can be inserted inside of the unit to keep the micro-SD card from being removed without opening the case. This accessory can be retrofitted into existing Connectbox version 6 units or later. But is best if pre-ordered with the unit.

Model P V8 can be ordered with additional:

Accessory 3

1 x Positive 18650 spacer & 1 x Negative 18650 spacer

This is an open source project and we appreciate your feedback and suggestions. Please visit: http://feedback.connectbox.technology/ and leave your comments. Anything posted here will automatically go directly to the developers for consideration.

ConnectBox Homepage: https://connectbox.technology/wp/

ConnectBox Development Source Code: https://github.com/ConnectBox